Paper 1 - LCR Chain Carrier

CQE-paper-01
CQECMPLX Formal Paper Series
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Construction Basis

Formalize Left-Center-Right readout as the smallest chain carrier that preserves a center while allowing two opposed boundary directions.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-01.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-01\paper_kernel_manifest.json
- body: production\papers\CQE-paper-01\PAPER-BODY.md
- source: production\papers\CQE-paper-01\SOURCE.md
- formal: production\papers\CQE-paper-01\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-01\02-CQE-tool\TOOL.md
- tool_runner: production\papers\CQE-paper-01\02-CQE-tool\run.py
- workbook: production\papers\CQE-paper-01\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-01.md

Paper 1 - LCR Chain Carrier

Abstract

This paper defines the first active object in the CQECMPLX sequence after the foreword and claim contract: the local Left-Center-Right carrier. The carrier is the ordered triple

```
(L, C, R)
```

where `C` is the active center selected by the observer's enumeration event, and `L` and `R` are the two boundary addresses read relative to that center.

The main theorem is deliberately small and foundational. Any local carrier that preserves one distinguished center and records two addressable boundary directions requires at least three positions. The ordered LCR triple realizes that lower bound. This proves that the LCR chart is the minimal carrier for the later papers' transport, correction, triality, boundary, and observer claims.

The paper also records the first physical bridge used by the suite: the `shell = 2` subspace of the binary LCR chart contains the three states

```
(1,1,0), (1,0,1), (0,1,1)
```

and the left-right side flip exchanges the two chiral states while fixing the middle state. This is the local structure later papers use as the single-forward-tape doublet carrier. Paper 1 proves the finite carrier facts directly and binds the stronger doublet interpretation to the downstream trace-2/Jordan and closure papers where its larger algebraic claims are completed.

Claims

Claim 1.1. The LCR carrier is the minimal ordered local carrier that preserves one center and two boundary addresses.

Claim 1.2. For the binary chart, there are exactly eight local states, and left-right reversal preserves `C`, is an involution, and partitions the states into four fixed states and two two-state reversal orbits.

Claim 1.3. Directional opposition is address-based, not value-based. Therefore the statement "opposed boundaries must have unequal values" is false. The counterexample is `(1,0,1)`, where `L = R = 1` while `L` and `R` remain distinct boundary addresses.

Claim 1.4. The `shell = 2` stratum supplies the first carried doublet interface: `(1,1,0)` and `(0,1,1)` are exchanged by left-right reversal, while `(1,0,1)` is fixed. The full physical reading of this interface is carried forward to the trace-2 and closure results in Papers 3 and later.

Predictions

The finite verifier for Paper 1 must return:

```
state count                        = 8
shell multiplicities               = 1, 3, 3, 1
left-right reversal preserves C     = true
left-right reversal is involutive   = true
fixed reversal states               = 4
```

```
two-state reversal orbits            = 2
counterexample to value inequality   = (1,0,1)
minimal address count                = 3
```

A verifier or reviewer that reports every `shell = 2` state as having unequal boundary values must fail.

Definitions

Let `A` be a finite alphabet. For the binary examples in this paper, `A = {0,1}`.

An **LCR state** over `A` is an ordered triple:

```
s = (L, C, R) in A^3
```

`C` is the distinguished center. `L` is the left boundary address read relative to `C`. `R` is the right boundary address read relative to `C`.

The **center projection** is:

```
pi_C(L,C,R) = C
```

The **left-right reversal** is:

```
rho(L,C,R) = (R,C,L)
```

The **binary shell** is:

```
shell(L,C,R) = L + C + R
```

The binary shell grades have multiplicity:

```
shell 0: 1 state
shell 1: 3 states
shell 2: 3 states
shell 3: 1 state
```

Directional opposition means:

```
address(L) != address(R)
```

Value inequality means:

```
value(L) != value(R)
```

Paper 1 requires directional opposition. It does not require value inequality in every state.

Test Protocol

The test is finite.

- Enumerate all triples in `{0,1}^3`.
- Compute `rho(s)` for every state.
- Check that `pi_C(rho(s)) = pi_C(s)` for every state.
- Check that `rho(rho(s)) = s` for every state.
- Count shell multiplicities.
- Count fixed states and two-state orbits under `rho`.
- Test the false boundary-value claim against `(1,0,1)`.
- Check that the address roles `{left_boundary, center, right_boundary}` have
cardinality three.

The production verifier is:

```
production/formal-papers/CQE-paper-01/verify_lcr_carrier.py
```

It emits:

```
production/formal-papers/CQE-paper-01/lcr_carrier_receipt.json
```

Results

The verifier returns `status = pass`.

The eight binary LCR states are:

```
(0,0,0) shell 0
(0,0,1) shell 1
(0,1,0) shell 1
(0,1,1) shell 2
(1,0,0) shell 1
(1,0,1) shell 2
(1,1,0) shell 2
(1,1,1) shell 3
```

The fixed states under left-right reversal are:

```
(0,0,0)
(0,1,0)
(1,0,1)
(1,1,1)
```

The two nontrivial reversal pairs are:

```
(0,0,1) <-> (1,0,0)
(0,1,1) <-> (1,1,0)
```

The counterexample to the false boundary-value claim is:

```
(1,0,1)
```

It has shell 2 and equal boundary values, but its left and right boundary addresses remain distinct.

Theorem 1.1 - Minimal LCR Carrier

Any ordered local carrier that preserves one distinguished center and records two addressable boundary directions requires at least three positions. The ordered triple `(L,C,R)` realizes this lower bound and is therefore minimal.

Proof

A carrier that preserves a distinguished center must contain an address for the center. A carrier that records two boundary directions relative to that center must contain one address for the left boundary and one address for the right boundary.

These three addresses cannot be identified. If the center address is identified with a boundary address, the center is no longer distinguished. If the two boundary addresses are identified, the carrier cannot distinguish left from right. Therefore any such carrier has at least three addresses.

The ordered triple `(L,C,R)` has exactly these three addresses. It preserves the center by `pi_C`, records both boundary directions, and admits the reversal operation `rho`. Therefore it attains the lower bound. QED.

Theorem 1.2 - Binary Reversal Inventory

In the binary LCR chart, left-right reversal preserves the center, is an involution, has four fixed states, and has two nontrivial two-state orbits.

Proof

For any state `(L,C,R)`:

```
pi_C(rho(L,C,R)) = pi_C(R,C,L) = C = pi_C(L,C,R)
```

and:

```
rho(rho(L,C,R)) = rho(R,C,L) = (L,C,R)
```

Thus reversal preserves the center and is an involution.

The fixed states satisfy `L = R`, giving:

```
(0,0,0), (0,1,0), (1,0,1), (1,1,1)
```

The remaining four states form two reversal pairs:

```
(0,0,1) <-> (1,0,0)
(0,1,1) <-> (1,1,0)
```

This exhausts the eight binary states. QED.

Doublet Interface

The `shell = 2` stratum is:

```
(1,1,0), (1,0,1), (0,1,1)
```

Left-right reversal maps:

```
(1,1,0) <-> (0,1,1)
(1,0,1) -> (1,0,1)
```

This gives the first local doublet interface used by the suite: two chiral states exchanged by side flip and one fixed pivot. The stronger statement that this interface is the single-forward-tape carrier for the SU(2)-style doublet is carried as `T_BIJECTIVE` in the surrounding evidence corpus and depends on the trace-2/Jordan registration and later closure papers. Paper 1 supplies the finite carrier and reversal facts those later claims require.

Falsifiers

The paper fails if any of the following occurs:

```
the binary state count is not 8
rho fails to preserve C
rho is not an involution
shell counts differ from 1,3,3,1
the fixed-state count is not 4
the nontrivial reversal-pair count is not 2
(1,0,1) is not recognized as a shell-2 equal-boundary counterexample
a two-address object is claimed to preserve one center and two distinct boundaries
```

Scope Boundary

Paper 1 proves the minimal carrier and the finite binary reversal inventory. It does not by itself prove the full Standard Model extension, the full `J_3(O)` registration, the full Rule 30 solve, or the complete spinor double-cover semantics. It supplies the first local carrier and the first finite doublet interface that later papers must preserve.

Evidence Bindings

Primary executable evidence:

```
production/formal-papers/CQE-paper-01/verify_lcr_carrier.py
production/formal-papers/CQE-paper-01/lcr_carrier_receipt.json
```

Supporting source evidence:

```
production/papers/CQE-paper-01/SOURCE.md
production/papers/CQE-paper-01/PAPER-BODY.md
D:/CQE_CMPLX/Claude-Codex-Memory/Claude work/CL-Paper-Evidence-DB/CL-CQE-Papers-01-05-C-Form-Chain.md
D:/CQE_CMPLX/CMPLX-R30-main/CATALOG/distilled_claims.json
```

Conclusion

Paper 1 establishes the local carrier that every later paper must respect. The result is not a metaphor and not a tool description. It is a finite structural theorem: one center plus two distinguishable boundary directions requires three ordered addresses, and the binary chart over those addresses has a fully enumerated reversal structure. Later papers may lift, repair, curve, or reinterpret that carrier, but they must preserve the accounting of `L`, `C`, `R`, and the observer-chosen center.